PADHAKU
Mathematics · Class 10 Mathematics

CHAPTER 3

Pair of Linear Equations in Two Variables

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text question · 3.1 Introduction

? **Akhila went to a fair. The number of times she played Hoopla is half the number of rides on the Giant Wheel. Each ride costs ₹3, a game of Hoopla ₹4, and she spent ₹20. How would you find the number of rides and the number of Hoopla games? Can we find the solutions of this pair of equations?**

✔ ANSWER

Let x = number of rides, y = number of Hoopla games.

y = x/2 ... (1) and 3x + 4y = 20 ... (2)

Put y = x/2 in (2): 3x + 4(x/2) = 20 → 3x + 2x = 20 → 5x = 20 → x = 4

y = 4/2 = 2

Yes, the pair can be solved (graphically or by algebra, as studied in this chapter).

Akhila had 4 rides and played Hoopla 2 times.

📝 Exercise 3.1

1 (i) **10 students of Class X took part in a Mathematics quiz. If the number of girls is 4 more than the number of boys, find the number of boys and girls who took part in the quiz (solve graphically).**

✔ ANSWER

Let number of girls = x, number of boys = y.

$x + y = 10$ ... (1) and $x - y = 4$ ... (2)

Points for $x + y = 10$: (0, 10), (5, 5), (10, 0)

Points for $x - y = 4$: (4, 0), (7, 3), (0, −4)

Plot both lines on the same graph paper. They intersect at the point (7, 3).

So $x = 7$, $y = 3$.

Girls = 7, Boys = 3

1 (ii) **5 pencils and 7 pens together cost ₹50, whereas 7 pencils and 5 pens together cost ₹46. Find the cost of one pencil and that of one pen (solve graphically).**

✔ ANSWER

Let cost of a pencil = ₹x and cost of a pen = ₹y.

$5x + 7y = 50$ ... (1) and $7x + 5y = 46$ ... (2)

Points for $5x + 7y = 50$: (10, 0), (3, 5), (−4, 10)

Points for $7x + 5y = 46$: (8, −2), (3, 5), (−2, 12)

Plot both lines. They intersect at the point (3, 5).

Cost of one pencil = ₹3, cost of one pen = ₹5

2 (i) **On comparing the ratios a_1/a_2, b_1/b_2 and c_1/c_2, find out whether the lines intersect at a point, are parallel or coincident: $5x - 4y + 8 = 0$; $7x + 6y - 9 = 0$**

✔ ANSWER

$a_1/a_2 = 5/7$, $b_1/b_2 = -4/6 = -2/3$

$5/7 \neq -2/3$, so $a_1/a_2 \neq b_1/b_2$.

The lines intersect at a point.

2 (ii) **$9x + 3y + 12 = 0; 18x + 6y + 24 = 0$**

✔ ANSWER

$a_1/a_2 = 9/18 = 1/2$, $b_1/b_2 = 3/6 = 1/2$, $c_1/c_2 = 12/24 = 1/2$

$a_1/a_2 = b_1/b_2 = c_1/c_2$

The lines are coincident.

2 (iii) **$6x - 3y + 10 = 0; 2x - y + 9 = 0$**

✔ ANSWER

$a_1/a_2 = 6/2 = 3$, $b_1/b_2 = -3/-1 = 3$, $c_1/c_2 = 10/9$

$a_1/a_2 = b_1/b_2 \neq c_1/c_2$

The lines are parallel.

3 (i) **On comparing the ratios a_1/a_2, b_1/b_2 and c_1/c_2, find out whether the pair is consistent or inconsistent: $3x + 2y = 5; 2x - 3y = 7$**

✔ ANSWER

$a_1/a_2 = 3/2$, $b_1/b_2 = 2/(-3) = -2/3$

$a_1/a_2 \neq b_1/b_2 \rightarrow$ unique solution.

Consistent

3 (ii) **$2x - 3y = 8; 4x - 6y = 9$**

✔ ANSWER

$a_1/a_2 = 2/4 = 1/2$, $b_1/b_2 = -3/-6 = 1/2$, $c_1/c_2 = 8/9$

$a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$ parallel lines, no solution.

Inconsistent

3 (iii) **(3/2)x + (5/3)y = 7; 9x − 10y = 14**

✔ ANSWER

$a_1/a_2 = (3/2)/9 = 3/18 = 1/6$, $b_1/b_2 = (5/3)/(-10) = -5/30 = -1/6$

$a_1/a_2 \neq b_1/b_2 \rightarrow$ unique solution.

Consistent

3 (iv) **5x − 3y = 11; −10x + 6y = −22**

✔ ANSWER

$a_1/a_2 = 5/(-10) = -1/2$, $b_1/b_2 = -3/6 = -1/2$, $c_1/c_2 = 11/(-22) = -1/2$

$a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$ coincident lines, infinitely many solutions.

Consistent (dependent)

3 (v) **(4/3)x + 2y = 8; 2x + 3y = 12**

✔ ANSWER

$a_1/a_2 = (4/3)/2 = 4/6 = 2/3$, $b_1/b_2 = 2/3$, $c_1/c_2 = 8/12 = 2/3$

$a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$ coincident lines, infinitely many solutions.

Consistent (dependent)

4 (i) **Consistent or inconsistent? If consistent, obtain the solution graphically: x + y = 5, 2x + 2y = 10**

✔ ANSWER

$a_1/a_2 = 1/2$, $b_1/b_2 = 1/2$, $c_1/c_2 = 5/10 = 1/2$ → all equal.

Consistent (dependent): the two lines coincide.

Points for x + y = 5: (0, 5), (2, 3), (5, 0); 2x + 2y = 10 gives the same points.

On the graph both equations give one and the same line, so every point on it is a solution.

Infinitely many solutions, given by y = 5 − x (x any real number).

4 (ii) **x − y = 8, 3x − 3y = 16**

✔ ANSWER

$a_1/a_2 = 1/3$, $b_1/b_2 = -1/-3 = 1/3$, $c_1/c_2 = 8/16 = 1/2$

$a_1/a_2 = b_1/b_2 \neq c_1/c_2$ → parallel lines.

Inconsistent (no solution).

4 (iii) **2x + y − 6 = 0, 4x − 2y − 4 = 0**

✔ ANSWER

$a_1/a_2 = 2/4 = 1/2$, $b_1/b_2 = 1/(-2) = -1/2 \rightarrow a_1/a_2 \neq b_1/b_2$

Consistent (unique solution).

Points for $2x + y = 6$: (0, 6), (2, 2), (3, 0)

Points for $4x - 2y = 4$: (1, 0), (2, 2), (0, −2)

Plot both lines; they intersect at (2, 2).

Solution: x = 2, y = 2

4 (iv) **$2x - 2y - 2 = 0, 4x - 4y - 5 = 0$**

✔ ANSWER

$a_1/a_2 = 2/4 = 1/2$, $b_1/b_2 = -2/-4 = 1/2$, $c_1/c_2 = -2/-5 = 2/5$

$a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$ parallel lines.

Inconsistent (no solution).

5 **Half the perimeter of a rectangular garden, whose length is 4 m more than its width, is 36 m. Find the dimensions of the garden.**

✔ ANSWER

Let length = x m, width = y m.

Half perimeter: $x + y = 36$ ... (1); length is 4 more: $x - y = 4$ ... (2)

Graphically: points for $x + y = 36$: (0, 36), (36, 0), (20, 16); for $x - y = 4$: (4, 0), (20, 16), (10, 6).

The lines meet at (20, 16).

(Check by adding (1) and (2): $2x = 40 \rightarrow x = 20$, $y = 16$.)

Length = 20 m, width = 16 m

6 (i) **Given $2x + 3y - 8 = 0$, write another linear equation in two variables such that the pair forms intersecting lines.**

✔ ANSWER

For intersecting lines we need $a_1/a_2 \neq b_1/b_2$.

Take **$3x + 2y - 7 = 0$** ($2/3 \neq 3/2$).

(Many other answers are possible.)

6 (ii) **... parallel lines**

✔ ANSWER

For parallel lines we need $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.

Take **$2x + 3y - 12 = 0$** ($2/2 = 3/3 \neq -8/-12$).

(Many other answers are possible.)

6 (iii) **... coincident lines**

✔ ANSWER

For coincident lines we need $a_1/a_2 = b_1/b_2 = c_1/c_2$.

Take **$4x + 6y - 16 = 0$** (each ratio = 1/2).

(Any multiple of the given equation works.)

7 **Draw the graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines**

and the x-axis, and shade the triangular region.

✔ ANSWER

$x - y + 1 = 0 \rightarrow y = x + 1$: points (0, 1), (−1, 0), (2, 3)

$3x + 2y - 12 = 0 \rightarrow y = (12 - 3x)/2$: points (4, 0), (0, 6), (2, 3)

Draw both lines. They intersect at (2, 3).

The first line meets the x-axis at (−1, 0); the second meets it at (4, 0).

Shade the triangle between the two lines and the x-axis.

Vertices of the triangle: (−1, 0), (4, 0) and (2, 3)

Exercise 3.2

1 (i) **Solve by the substitution method: $x + y = 14$; $x - y = 4$**

✔ ANSWER

From (2): $x = y + 4$

Put in (1): $y + 4 + y = 14 \rightarrow 2y = 10 \rightarrow y = 5$

$x = 5 + 4 = 9$

$x = 9, y = 5$

1 (ii) **$s - t = 3$; $s/3 + t/2 = 6$**

✔ ANSWER

From (1): s = t + 3

(2) × 6: 2s + 3t = 36

Put s = t + 3: 2(t + 3) + 3t = 36 → 5t + 6 = 36 → 5t = 30 → t = 6

s = 6 + 3 = 9

s = 9, t = 6

1 (iii) **3x − y = 3; 9x − 3y = 9**

✔ ANSWER

From (1): y = 3x − 3

Put in (2): 9x − 3(3x − 3) = 9 → 9x − 9x + 9 = 9 → 9 = 9, which is true for every x.

So the two equations represent the same line (the second is 3 times the first).

Infinitely many solutions: y = 3x − 3, where x is any real number.

1 (iv) **0.2x + 0.3y = 1.3; 0.4x + 0.5y = 2.3**

✔ ANSWER

Multiply both by 10: 2x + 3y = 13 ... (1), 4x + 5y = 23 ... (2)

From (1): x = (13 − 3y)/2

Put in (2): 4(13 − 3y)/2 + 5y = 23 → 26 − 6y + 5y = 23 → y = 3

x = (13 − 9)/2 = 2

x = 2, y = 3

1 (v) **√2 x + √3 y = 0; √3 x − √8 y = 0**

✔ ANSWER

From (1): x = −(√3/√2) y

Put in (2): √3 × (−√3/√2) y − 2√2 y = 0 → −(3/√2) y − 2√2 y = 0

Multiply by √2: −3y − 4y = 0 → −7y = 0 → y = 0

x = −(√3/√2) × 0 = 0

x = 0, y = 0

1 (vi) **3x/2 − 5y/3 = −2; x/3 + y/2 = 13/6**

✔ ANSWER

(1) × 6: 9x − 10y = −12 ... (3); (2) × 6: 2x + 3y = 13 ... (4)

From (4): x = (13 − 3y)/2

Put in (3): 9(13 − 3y)/2 − 10y = −12 → 117 − 27y − 20y = −24 → 47y = 141 → y = 3

x = (13 − 9)/2 = 2

x = 2, y = 3

2 **Solve 2x + 3y = 11 and 2x − 4y = −24 and hence find the value of m for which y = mx + 3.**

✔ ANSWER

From (1): x = (11 − 3y)/2

Put in (2): 11 − 3y − 4y = −24 → −7y = −35 → y = 5

x = (11 − 15)/2 = −2

x = −2, y = 5

Now y = mx + 3 → 5 = m(−2) + 3 → −2m = 2 → **m = −1**

3 (i) **The difference between two numbers is 26 and one number is three times the other. Find them.**

✔ ANSWER

Let the numbers be x and y (x > y).

x − y = 26 ... (1), x = 3y ... (2)

Put (2) in (1): 3y − y = 26 → 2y = 26 → y = 13

x = 3 × 13 = 39

The numbers are 39 and 13.

3 (ii) **The larger of two supplementary angles exceeds the smaller by 18 degrees. Find them.**

✔ ANSWER

Let the angles be x° (larger) and y° (smaller).

x + y = 180 ... (1), x − y = 18 ... (2)

From (2): x = y + 18. Put in (1): y + 18 + y = 180 → 2y = 162 → y = 81

x = 81 + 18 = 99

The angles are 99° and 81°.

3 (iii) **The coach of a cricket team buys 7 bats and 6 balls for ₹3800. Later, she buys 3**

bats and 5 balls for ₹1750. Find the cost of each bat and each ball.

✔ ANSWER

Let cost of a bat = ₹x, cost of a ball = ₹y.

7x + 6y = 3800 ... (1), 3x + 5y = 1750 ... (2)

From (2): x = (1750 − 5y)/3

Put in (1): 7(1750 − 5y)/3 + 6y = 3800 → 12250 − 35y + 18y = 11400 → 17y = 850 → y = 50

x = (1750 − 250)/3 = 500

Bat = ₹500, ball = ₹50

3 (iv) **Taxi charges consist of a fixed charge plus a charge per km. For 10 km the charge is ₹105, for 15 km it is ₹155. What are the fixed charge and the charge per km? How much for 25 km?**

✔ ANSWER

Let fixed charge = ₹x, charge per km = ₹y.

x + 10y = 105 ... (1), x + 15y = 155 ... (2)

From (1): x = 105 − 10y. Put in (2): 105 − 10y + 15y = 155 → 5y = 50 → y = 10

x = 105 − 100 = 5

Fixed charge = ₹5, charge per km = ₹10

For 25 km: 5 + 10 × 25 = **₹255**

3 (v) **A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is**

added to both, it becomes 5/6. Find the fraction.

✔ ANSWER

Let the fraction be x/y.

$(x + 2)/(y + 2) = 9/11 \rightarrow 11x + 22 = 9y + 18 \rightarrow 11x - 9y + 4 = 0$ ... (1)

$(x + 3)/(y + 3) = 5/6 \rightarrow 6x + 18 = 5y + 15 \rightarrow 6x - 5y + 3 = 0$ ... (2)

From (2): $y = (6x + 3)/5$

Put in (1): $11x - 9(6x + 3)/5 + 4 = 0 \rightarrow 55x - 54x - 27 + 20 = 0 \rightarrow x = 7$

$y = (42 + 3)/5 = 9$

The fraction is 7/9.

3 (vi) **Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. What are their present ages?**

✔ ANSWER

Let Jacob's age = x years, son's age = y years.

$x + 5 = 3(y + 5) \rightarrow x - 3y = 10$ ... (1)

$x - 5 = 7(y - 5) \rightarrow x - 7y = -30$ ... (2)

From (1): $x = 3y + 10$. Put in (2): $3y + 10 - 7y = -30 \rightarrow -4y = -40 \rightarrow y = 10$

$x = 30 + 10 = 40$

Jacob is 40 years old, his son is 10 years old.

In-text questions · 3.3.2 Elimination Method (Remarks after Example 8)

1 **In Example 8 (9x − 4y = 2000, 7x − 3y = 2000) we eliminated y. We could also have eliminated x. Try doing it that way.**

✔ ANSWER

(1) × 7: 63x − 28y = 14000; (2) × 9: 63x − 27y = 18000

Subtract: (63x − 63x) + (−27y + 28y) = 18000 − 14000 → y = 4000

Put y = 4000 in (1): 9x − 16000 = 2000 → 9x = 18000 → x = 2000

Incomes = 9x and 7x: **₹18,000 and ₹14,000** (same answer as before).

2 **You could also have used the substitution or graphical method to solve this problem. Try doing so, and see which method is more convenient.**

💡 SAMPLE ANSWER

Substitution: from (2), x = (2000 + 3y)/7. Put in (1): 9(2000 + 3y)/7 − 4y = 2000 → 18000 + 27y − 28y = 14000 → y = 4000, x = 2000.

Graphical: the lines 9x − 4y = 2000 and 7x − 3y = 2000 meet at (2000, 4000), but the numbers are very large, so the graph is hard to draw and read accurately.

The elimination method is the most convenient here (no fractions, no large graph).

📝 Exercise 3.3

1 (i) **Solve by the elimination method and the substitution method: x + y = 5 and 2x − 3y = 4**

✔ ANSWER

Elimination: (1) × 3: 3x + 3y = 15; add to (2): 5x = 19 → x = 19/5

y = 5 − 19/5 = 6/5

Substitution: y = 5 − x; 2x − 3(5 − x) = 4 → 5x = 19 → x = 19/5, y = 6/5

x = 19/5, y = 6/5

1 (ii) **3x + 4y = 10 and 2x − 2y = 2**

✔ ANSWER

Elimination: (2) × 2: 4x − 4y = 4; add to (1): 7x = 14 → x = 2

2(2) − 2y = 2 → y = 1

Substitution: from (2) x = y + 1; 3(y + 1) + 4y = 10 → 7y = 7 → y = 1, x = 2

x = 2, y = 1

1 (iii) **3x − 5y − 4 = 0 and 9x = 2y + 7**

✔ ANSWER

Write as $3x - 5y = 4$ ... (1), $9x - 2y = 7$ ... (2)

Elimination: (1) × 3: $9x - 15y = 12$; subtract from (2): $13y = -5 \rightarrow y = -5/13$

$3x = 4 + 5(-5/13) = 4 - 25/13 = 27/13 \rightarrow x = 9/13$

Substitution: $x = (4 + 5y)/3$; $9(4 + 5y)/3 - 2y = 7 \rightarrow 12 + 15y - 2y = 7 \rightarrow 13y = -5 \rightarrow y = -5/13$, $x = 9/13$

$x = 9/13, y = -5/13$

1 (iv) **$x/2 + 2y/3 = -1$ and $x - y/3 = 3$**

✔ ANSWER

(1) × 6: $3x + 4y = -6$ ... (3); (2) × 3: $3x - y = 9$ ... (4)

Elimination: (3) − (4): $5y = -15 \rightarrow y = -3$

$3x - (-3) = 9 \rightarrow 3x = 6 \rightarrow x = 2$

Substitution: from (4) $y = 3x - 9$; $3x + 4(3x - 9) = -6 \rightarrow 15x = 30 \rightarrow x = 2$, $y = -3$

$x = 2, y = -3$

2 (i) **If we add 1 to the numerator and subtract 1 from the denominator, a fraction reduces to 1. It becomes 1/2 if we only add 1 to the denominator. What is the fraction?**

✔ ANSWER

Let the fraction be x/y.

$(x + 1)/(y - 1) = 1 \rightarrow x + 1 = y - 1 \rightarrow x - y = -2$ ... (1)

$x/(y + 1) = 1/2 \rightarrow 2x = y + 1 \rightarrow 2x - y = 1$ ... (2)

(2) − (1): $x = 3$; then $y = x + 2 = 5$

The fraction is 3/5.

2 (ii) **Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu?**

✔ ANSWER

Let Nuri's age = x years, Sonu's age = y years.

$x - 5 = 3(y - 5) \rightarrow x - 3y = -10$ ... (1)

$x + 10 = 2(y + 10) \rightarrow x - 2y = 10$ ... (2)

(2) − (1): $y = 20$; $x = 10 + 2(20) = 50$

Nuri is 50 years old, Sonu is 20 years old.

2 (iii) **The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the digits. Find the number.**

✔ ANSWER

Let tens digit = x, units digit = y. Number = 10x + y; reversed number = 10y + x.

$x + y = 9$ ... (1)

$9(10x + y) = 2(10y + x) \rightarrow 90x + 9y = 20y + 2x \rightarrow 88x - 11y = 0 \rightarrow 8x - y = 0$ ... (2)

(1) + (2): $9x = 9 \rightarrow x = 1$; $y = 8$

The number is 18. (Check: $9 \times 18 = 162 = 2 \times 81$)

2 (iv) **Meena went to a bank to withdraw ₹2000. She asked for ₹50 and ₹100 notes only and got 25 notes in all. How many notes of each kind did she receive?**

✔ ANSWER

Let number of ₹50 notes = x, ₹100 notes = y.

x + y = 25 ... (1)

50x + 100y = 2000 → x + 2y = 40 ... (2)

(2) − (1): y = 15; x = 25 − 15 = 10

10 notes of ₹50 and 15 notes of ₹100

2 (v) **A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid ₹27 for a book kept for seven days, Susy paid ₹21 for five days. Find the fixed charge and the charge for each extra day.**

✔ ANSWER

Let fixed charge = ₹x, charge per extra day = ₹y.

7 days = 3 + 4 extra: x + 4y = 27 ... (1)

5 days = 3 + 2 extra: x + 2y = 21 ... (2)

(1) − (2): 2y = 6 → y = 3; x = 21 − 6 = 15

Fixed charge = ₹15, charge per extra day = ₹3

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**